

Software Engineering Lab 1: Requirements Engineering

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1 Scope of the Engine

The search engine allows users to efficiently search for information on the web with both basic and advanced capabilities.

Main Functions

- **Searching and filtering:** Users can perform general searches or use quotation marks (‘ ’) to find exact terms instantly. Results can be filtered based on relevance, type (news, websites, media), or safety rating.
- **Reporting system:** Users can report scams, inappropriate data, or unsafe websites.
- **Ad control:** Option to block or minimize unwanted ads for a cleaner experience.
- **Administrative tools:** Admins can review user reports, verify flagged content, and take action against harmful or illegal material.

Extra Features

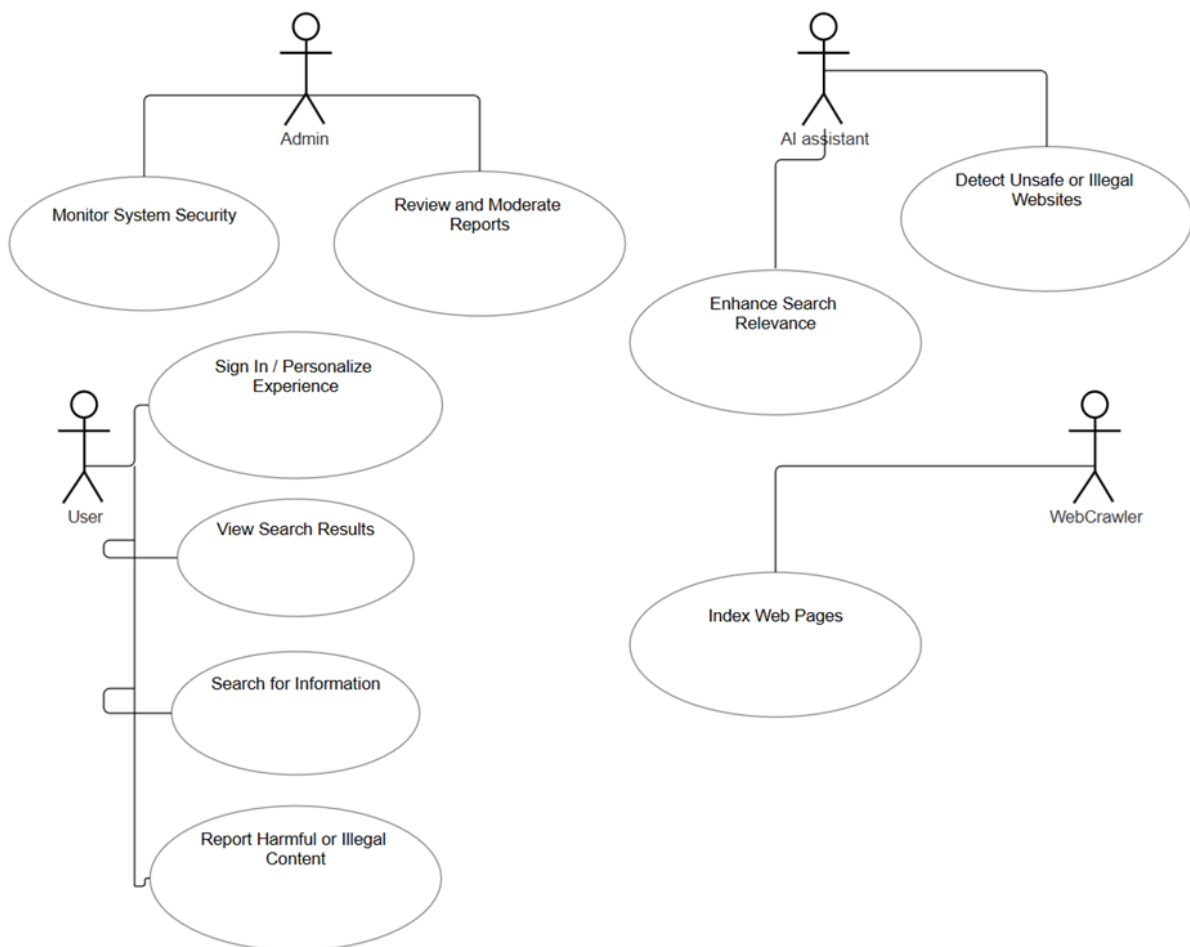
- Voice search via speech commands.
- Auto-correct and “Did you mean” suggestions for typos.
- Favorites section for quick access to frequently used apps or sites.
- Personalized news feed with topic/source preferences.
- Theme customization for interface personalization.

2 Actors

- **User:** Performs searches, manages preferences, and reports inappropriate content. Signing in enables personalization and security tracking.
- **Admin/Moderator:** Reviews reports, verifies flagged data, ensures safety compliance, and maintains database integrity.

- **AI Assistant:** Supports faster data retrieval, enhances ranking relevance, and flags unsafe or malicious content.
- **WebCrawler:** Collects and indexes web data for search retrieval.
- **Website Owner:** Updates indexed data and ensures compliance with content policies.

3 Use Case Diagram



4 Use Case Descriptions

4.1 UC1 – Search for Information

- **Initiating Actor:** User
- **Actor’s Goal:** Find relevant information on the web.
- **Participating Actors:** AI Assistant, WebCrawler
- **Preconditions:** System is online and has an updated index.
- **Postconditions:** Search results are displayed to the user.

Main Success Scenario:

1. User enters a search query.
2. System sends query to AI Assistant for processing.
3. AI Assistant interprets intent and retrieves matching indexed pages.
4. System ranks and displays the most relevant results.

Alternate Scenario:

1. No results found.
2. System suggests “Did you mean...?” options for possible typos or related terms.
3. System displays “No matches found” and suggests similar queries.

4.2 UC2 – View Search Results

- **Initiating Actor:** User
- **Actor’s Goal:** Read or access the displayed results.
- **Participating Actors:** AI Assistant
- **Preconditions:** Query completed successfully.
- **Postconditions:** User may open or interact with result links.

Main Success Scenario:

1. User views the result list.
2. User selects a desired webpage link.
3. System redirects to external webpage or cached copy.

Alternate Scenario:

1. Link unavailable.
2. System offers cached or alternative version.

4.3 UC3 – Sign In / Personalize Experience

- **Initiating Actor:** User
- **Actor's Goal:** Access personalized search features.
- **Participating Actors:** System
- **Preconditions:** User has account credentials.
- **Postconditions:** Personalized settings are applied.

Main Success Scenario:

1. User selects “Sign In”.
2. System authenticates credentials.
3. Personalized features (history, preferences) are loaded.

Alternate Scenario:

1. Invalid credentials.
2. System shows error and offers password reset.

4.4 UC4 – Report Harmful or Illegal Content

- **Initiating Actor:** User
- **Actor's Goal:** Notify the system about suspicious content.
- **Participating Actors:** Admin/Moderator, AI Assistant
- **Preconditions:** User must be viewing a result.
- **Postconditions:** Report is logged for review.

Main Success Scenario:

1. User clicks “Report” button.
2. System sends report to moderator.
3. Moderator receives flagged content details.

Alternate Scenario:

1. Report submission fails.
2. System prompts user to retry.

4.5 UC5 – Review and Moderate Reports

- **Initiating Actor:** Admin/Moderator
- **Actor's Goal:** Review user reports and ensure content safety.
- **Participating Actors:** AI Assistant
- **Preconditions:** At least one report exists.
- **Postconditions:** Action (approve, remove, or ignore) recorded.

Main Success Scenario:

1. Admin logs into moderation dashboard.
2. System lists pending reports.
3. Admin investigates and takes appropriate action (ban, suspend, or ignore).

Alternate Scenario:

1. Report lacks data.
2. System requests additional information or marks as incomplete.

4.6 UC6 – Monitor System Security

- **Initiating Actor:** Admin/Moderator
- **Actor's Goal:** Maintain integrity and availability of the search engine.
- **Participating Actors:** AI Assistant
- **Preconditions:** System is operational.
- **Postconditions:** Potential security threats are logged or mitigated.

Main Success Scenario:

1. Admin views security dashboard.
2. AI Assistant displays alerts and trends.
3. Admin takes action on detected threats.

Alternate Scenario:

1. Dashboard unavailable.
2. Admin switches to backup monitoring mode.